

Evaluating design-stage manufacturability decisions against the resolution of production

Hüseyin Tayyer Canseven^{1*}

^{1*}Department of Electrical Engineering, LUT School of Energy Systems,
LUT University, Yliopistonkatu 34, Lappeenranta, 53850, Finland.

Corresponding author(s). E-mail(s): huseyin.canseven@lut.fi;

Abstract

Design-stage manufacturability decisions increasingly rest on simulation and machine learning, yet whether a prediction is fit to decide is rarely evaluated against production. A framework is proposed whose unit, the production floor, is the difference that drift and scatter produce between two batches of one design within a run. A rule returning meets, fails or trial is characterised by a decisive distance in floors, beyond which at least 95 % of its verdicts are correct, and a safe distance, beyond which at most 5 % are wrong, for each relation of a new design to the produced evidence: a variant, a new process setting or a new family. A procedure turns them into a guard band for single verdicts. Thirteen rules were evaluated at a 95 % conformance level on open data of 18 deep-drawn design–process alternatives of 500 parts each, mostly process settings on one tool, with matched simulations; one floor of draw-in is 0.05 mm. With a produced sibling the nearest produced setting decided within 3.4 floors; for a new process setting no rule decided closer than 8.5 floors; in two transfers to a new family none was safe closer than 16. Within a family the choice of rule mattered more than the relation, and requirements at a performance index of 1.33 lay inside every decisive distance. With six to nine produced settings per fit, the simulation, as used, helped only across families, and learned models added abstention rather than accuracy. The framework gives evaluations of AI-based manufacturability support a production-referenced standard.

Keywords: design for manufacturing, machine learning, design decision making, verification and validation, operating characteristic, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming, as in most processes, the decision rests on a finite-element simulation (Banabic, 2010): the team simulates the alternative, compares the predicted characteristics with the requirements and releases the alternative, changes it or sends it to a physical trial. Simulation has moved such decisions forward in the design process (Thomke, 1998), and machine learning makes the loop faster and broader: surrogates replace simulations, models learn manufacturability from earlier designs, and checks are automated (Yan et al, 2026; Ghadai et al, 2018; Zhang et al, 2018). Neither has changed the question the team has to answer: *can this prediction be relied on to decide?* A simulation that is right on average can be wrong for the alternative at hand, and a model learned on earlier products may be asked about a design unlike any of them. Whitney argued in this journal that mechanical design cannot be verified by simulation the way integrated circuits are (Whitney, 1996); production variation, with which development organisations still contend (Thornton et al, 2000), adds a further reason. Whether a simulation-based decision is credible is in the end a property of production, and production evidence is what is scarce at the design stage.

This paper treats the design-stage manufacturability decision itself as the object of study and asks three questions. How can the fitness of a design-stage prediction for a manufacturability decision be expressed on a common production-referenced scale, for any predictor and characteristic (RQ1)? How does that fitness depend on the production evidence available, on its amount and on how the new design relates to what has been produced (RQ2)? And what do simulation and machine learning contribute to such a decision once production evidence exists (RQ3)? The questions matter for decision-based design (Hazelrigg, 1998; Lewis et al, 2006), for the margins that designers keep between predictions and requirements (Eckert et al, 2019; Brahma and Wynn, 2020) and for the planning of verification and testing (Thomke and Bell, 2001; Salado and Kannan, 2018; Tahera et al, 2019): each needs to know how far a prediction can be relied on before a test is cheaper than a decision.

The framework proposed here has the *production floor* as its unit: the difference that drift and scatter produce between two batches of consecutive parts of one design within a run, a floor in the sense of a noise floor rather than of the shop floor. A rule that turns the design-stage evidence into a verdict (the alternative *meets* the requirement, *fails* it, or goes to a physical *trial*) is characterised by two operating characteristics in floors, its *decisive distance* and its *safe distance*, for each *evidence relation* between a new design and the produced alternatives, and a procedure turns them into a guard band on the margin of a single verdict. The decisions in scope are those of a team that has produced variants of a design family: the release of a further variant or process setting in process planning and tryout, and the transfer of production knowledge to a new family; early product design without production evidence of the family is covered only by the new-family relation.

The framework is evaluated on open data that make this possible: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design-process alternatives of

500 consecutive parts each and scanned (Baum and Heinzelmann, 2026), with finite-element simulations of the same process (Baum and Heinzelmann, 2025). Within each of two cup geometries the alternatives differ in blank-holder force and lubrication on one tool, so two of the three relations concern process settings, and the two geometries give two transfers between families. Each alternative is judged as a new design, and the verdicts of thirteen rules at a 95 % conformance level are compared with what its 500 parts did. In the terms of the design research methodology (Blessing and Chakrabarti, 2009), this is a prescriptive study with an initial support evaluation: it shows whether the framework discriminates between rules and evidence situations, not whether design teams decide better with it, and its findings on one dataset are hypotheses for confirmation.

The contributions are the framework, with the floor positioned as a precision limit of batch centres and the distances as operating characteristics with an exact estimator (Section 3); an evaluation design that separates the evidence relations by grouped cross-validation (Roberts et al, 2017) and isolates what the simulation adds by evaluating every calibrated rule with and without it (Section 4); the empirical answers for deep drawing (Section 5); and guidance and a reporting standard for evaluating simulation- and AI-based manufacturability support elsewhere (Section 6). Section 2 positions the work and Section 7 concludes; Appendix A summarises the notation, and the Electronic Supplementary Material (ESM) holds further results.

2 Background and related work

2.1 Manufacturability evaluation and machine learning

Design for manufacturing (DFM) codifies what a process can make into rules, guidelines and cost models applied at the design stage (Boothroyd et al, 2010), and quantitative manufacturability models have been used in this journal to drive design decisions (La Trobe-Bateman and Wild, 2003). Process simulation predicts the outcome of a specific alternative, and machine learning learns manufacturability from CAD models (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019), from process data and from simulated design spaces (Yan et al, 2026; Ballegeer and Benoit, 2026), and generative models propose designs directly (Regenwetter et al, 2022). Work in this journal has used data and models to choose processes from shape (Ben Slama et al, 2024) and shown that measures left out of an alternative-selection problem change the choice (Eaton et al, 2026). Transfer between processes, machines and product families is a recurring difficulty (Safdar et al, 2024). Learned manufacturability models are typically evaluated by held-out accuracy or classification scores on labelled designs (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019), not by the decisions they support against the production of the very alternatives they are asked about; open datasets that would allow more are recognised as contributions in their own right (Ahmed et al, 2025).

2.2 Design decisions, design types and production evidence

Decision-based design casts design as a choice between alternatives with uncertain outcomes (Hazelrigg, 1998; Lewis et al, 2006). How much a team knows about a new alternative depends on the kind of design: variant design changes parameters of a proven solution, adaptive design changes parts of it and original design creates a new one (Pahl et al, 2007), and product-family and platform design organise these changes across products (Simpson et al, 2006; Jiao et al, 2007). The evidence relations of this paper are their counterparts for production evidence: a new variant of a produced setting, a new process setting of a produced design, and a new family. Production knowledge enters design through process-capability data (Thornton, 2004; Tata and Thornton, 1999), key characteristics (Thornton, 1999) and tolerance analysis (Chase and Parkinson, 1991); a rule that returns the capability of the nearest produced setting is a capability-database look-up. Where the uncertainty of a prediction is not resolved it is absorbed by margins (Eckert et al, 2019; Brahma and Wynn, 2020), and the part of a margin that covers uncertainty is distinct from the part that absorbs change (Eckert and Isaksson, 2017). Testing and prototyping are the alternative to a margin: when to test, how many alternatives to test and in which order has been modelled as a trade between the cost of tests and the cost of errors (Thomke, 1998; Thomke and Bell, 2001; Salado and Kannan, 2018), with testing and prototyping strategies studied as design activities (Tahera et al, 2019; Camburn et al, 2017). When upstream information is fit to act on is the question of overlapped development and preliminary information (Krishnan et al, 1997; Terwiesch et al, 2002), and set-based concurrent engineering codifies production knowledge as feasibility limits for new designs (Sobek et al, 1999). All of these assume a measure of how far a design-stage prediction can be relied on; the framework supplies one in a unit set by production.

2.3 Simulation credibility, calibration and validation

Verification and validation ask whether a model is accurate enough for its intended use (Oberkampf and Roy, 2010), formalised in risk-informed credibility assessment, which scales the required evidence with the consequence of a wrong decision (ASME, 2018). The distinction between the validation domain and the application domain of a model (Oberkampf and Roy, 2010) is the question that the evidence relation asks of a design. Bayesian calibration adds a discrepancy model and calibration parameters to a simulator (Kennedy and O'Hagan, 2001), multi-fidelity models scale a cheap model to an expensive one (Kennedy and O'Hagan, 2000), and omitting the discrepancy biases the parameters (Brynjarsdóttir and O'Hagan, 2014). Where a predictive model may be relied on at all is the question of its valid input domain (Malak and Paredis, 2010). Conformal prediction gives distribution-free intervals with a finite-sample coverage guarantee under exchangeability (Lei et al, 2018; Barber et al, 2021), including for quantiles (Romano et al, 2019), under covariate shift (Tibshirani et al, 2019) and for the control of decision risks (Angelopoulos et al, 2024), and it has begun to appear in engineering design (Zong et al, 2026). In sheet-metal forming the accuracy of finite-element predictions has long been studied (Wagoner et al, 2013), with friction (Hol et al, 2012) and the unloading behaviour of the material (Yoshida and

Uemori, 2002) among the main sources of error. Simulation datasets of deep drawing have been released for surrogate and transfer learning (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025), the gap between synthetic and real data has been identified as the obstacle to industrial use (Baum et al, 2026b), and the deviation between simulation and parts has been decomposed statistically (Baum et al, 2026a). These methods produce intervals and accuracy statements; how close to a requirement such an interval can be relied on to decide, and how much production evidence that takes, remains open.

2.4 Decisions with abstention and their operating characteristics

In conformity assessment, a measured value proves conformance, proves non-conformance or leaves the decision open, with guard bands set by the measurement uncertainty and the risks of false acceptance and false rejection as the measures of a decision rule (ISO, 2017; Joint Committee for Guides in Metrology, 2012). In selection procedures and acceptance sampling the operating characteristic gives the probability of a decision as a function of the true quality, with an indifference zone inside which no decision is required (Bechhofer, 1954). A classifier with a reject option trades errors against abstentions (Chow, 1970), summarised by risk–coverage curves (Geifman and El-Yaniv, 2017); learning to defer passes difficult cases to an expert (Madras et al, 2018); and appropriate reliance on automation requires that its reliability be visible to its users (Lee and See, 2004), which matters for design teams working with AI (Zhang et al, 2021). The decisive and safe distances are operating characteristics of this kind for design-stage manufacturability decisions, and the guard band of the procedure is the design-stage counterpart of a conformity-assessment guard band.

2.5 Production variation and measurement

Statistical quality control separates short-term and long-term variation and compares them with tolerances through capability and performance indices (Montgomery, 2019), with release criteria such as a performance index of at least 1.33 (Automotive Industry Action Group, 2006) and time-dependent process models for drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5 % (ISO, 2025), and measurement systems analysis separates repeatability from stability and decides whether a gauge can resolve a tolerance (Automotive Industry Action Group, 2010). The production floor is a repeatability-type limit of batch centres under within-run conditions; what is new is its use as the unit of a design decision rule. Table 1 places the framework among these approaches: it does not replace calibration or conformance prediction but scores them, as decision rules, against production in a common unit.

Table 1 Positioning among methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
Design margins (Eckert et al, 2019)	experience, analysis	buffer to the requirement	rarely	no
V&V, credibility assessment (Oberkampff and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy in physical units; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O’Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage; per decision possible on the measured item	yes
Conformity assessment, OC curves (ISO, 2017; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks		yes
Selective classification (Chow, 1970; Geifman and El-Yaniv, 2017)	labelled data	class or reject; risk–coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This paper	simulations, produced alternatives	verdict; decisive and safe distance in floors by evidence relation	yes, per verdict	yes

3 The framework

3.1 The decision problem and its scope

A design alternative a is a combination of product geometry and process settings that is produced as a series of parts. A quality characteristic c varies over the parts of a series. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet: alternative a is *adequate* for c when the p -quantile of its parts satisfies $q_p(a, c) \leq R$. Lower or two-sided limits follow by a change of sign or by the absolute deviation from the target; a tolerance of $\pm t$ on a wall angle about its design angle, for example, is the limit $R = t$ on the absolute deviation. At the design stage q_p is unknown. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_p and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \quad (1)$$

The three verdicts mirror the decision rules of conformity assessment, in which uncertainty creates a zone where neither conformance nor non-conformance can be proven (ISO, 2017); here the uncertainty is that of the design-stage prediction. A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of simulations of a and of the production record of other alternatives, the *calibration alternatives*; it is used by the process planner or tryout engineer for a further variant or process setting of an existing tool, at tool tryout or before production part approval, and by the product designer for a new family. The demonstration decides on $p = 0.95$, which for a normal characteristic corresponds to a performance index of about 0.55. Release criteria of series production, a performance index of 1.33 or more (Automotive Industry Action Group, 2006), concern nonconforming fractions in parts per million that a series of 500 parts cannot estimate without a parametric tail; the framework reaches them through requirements anchored in capability (Section 5.8).

3.2 Production floor, effect-to-scatter ratio and minimum resolvable change

Production evidence comes as series of consecutive parts, each cut into batches of B consecutive parts, and each batch is summarised by a robust centre m (here the 10 % trimmed mean). The production floor of characteristic c is

$$F_c = Q_\alpha \{ |m_i - m_j| : i < j \text{ batches of the same alternative} \}, \quad (2)$$

the α -quantile of the difference between two batches of one design. Under a stationary normal model with a standard deviation σ_b between batch means and σ_w between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_b^2 + \sigma_w^2/B}$ at $\alpha = 0.95$: the repeatability limit of precision experiments (ISO, 2025, 1994), applied to batch centres under within-run conditions. The reference protocol of this paper is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$, with per-geometry floors where geometries differ. The floor rests on three assumptions: batches of one design differ only by drift and scatter within a run (A1); the runs from which it is estimated are like those of future production (A2); and the measuring system is repeatable and its resolution small against the floor (A3, checked in Section 5.1).

Four properties matter for its use. It has the unit of the characteristic, so distances in floors are on a common scale for all characteristics and predictors, although equal distances do not mean equal nonconforming fractions, because the ratio of the floor to the scatter of the parts differs between characteristics (Table 3). It is a property of a production protocol: under drift it grows with the time between the batches compared, and it shrinks with B ; σ_b , σ_w and the floor at every lag are reported so that it can be recomputed for another protocol (ESM Tables S3 and S4). It contains whatever drifts between batches, the process and the measuring system alike, so without a reference artefact scanned through the series it is a production-and-measurement floor. And it excludes the variation between runs, which makes it a lower bound of the variation of series production. For a new family, whose floor is unknown before it is produced, the floor of the produced family is used. Because the floor resolves batch centres while

the requirement concerns a quantile, the reproducibility of the quantile is reported alongside.

Two alternatives a and b are resolvable for c when their effect-to-scatter ratio

$$\text{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b| / F_c \quad (3)$$

is at least one, $\bar{m}$ being the centre of all parts of an alternative; this is a practical threshold, not a test. For a continuous design or process parameter x with a local sensitivity $s_c = \partial \bar{m} / \partial x$, the minimum resolvable change is $F_c / |s_c|$: a change of x by less moves the centre of the characteristic by less than two batches of one design differ. It is local, a statement about centres, and a change below it can still move the tail of the distribution.

3.3 Decision rules

Thirteen rules span what a design team can do with its evidence (Table 2); each is defined for any set of calibration alternatives, and the first seven carry the results in the main text. The nominal simulation (M0) uses the simulation alone. The bias-corrected simulation (M1) adds the median offset between the calibration alternatives' q_p and their nominal simulation; the scaled simulation (M1s) fits $q_p = a + b s_0$ to them by least squares, a linear multi-fidelity correction (Kennedy and O'Hagan, 2000) that rescales a simulated trend of the wrong size. The envelope rule (M2) adds the median offset and the largest absolute residual of the calibration alternatives to the upper end of the simulation envelope over plausible incoming conditions. M5 models the offset between q_p and the envelope as a Gaussian process (Rasmussen and Williams, 2006) with a 90 % predictive interval, whose noise variance is bounded below by the variance between the produced series of one geometry and force, so that a new series of a produced setting is not predicted more precisely than produced series reproduce each other; M5n is the same process of q_p without the simulation. The nearest produced setting (NN) is a capability-database look-up that uses neither simulation nor learning. The six further rules include a friction-calibrated simulation (Kennedy and O'Hagan, 2001) (Mc) and part-level gradient boosting (Friedman, 2001) with a jackknife+ interval (Barber et al, 2021), with and without the simulation (M3, M3n). The nominal simulation and the envelope do not depend on the lubrication pattern, so a lubrication variant of a produced setting gives M0, M1, M1s, M2 and M5 no simulated information that distinguishes it from its siblings.

The margins are labelled by what they guarantee. With n calibration alternatives, the largest-residual margin covers a new exchangeable residual with probability $n/(n+1)$ for a fixed offset and at least $(n-1)/(n+1)$ for an estimated one, because the new residual is then covered whenever it falls within the range of the calibration residuals; it is the split-conformal quantile (Lei et al, 2018) at the highest attainable level, not a 90 % interval. In general, a distribution-free interval cannot certify a miss rate below $1/(n+1)$, so a miss rate of 5 % needs at least 19 produced alternatives in the relevant relation and one of 10 % needs nine. Under exchangeability the jackknife+ interval covers with probability at least 0.8 at the level of 0.9 used here (Barber et al, 2021), and the Gaussian-process interval is model-based. The level is a decision variable that

Table 2 The decision rules; the first seven carry the results in the main text

Rule	Prediction of q_p	Interval	Production evidence
M0 nominal simulation	simulation at nominal sheet thickness and friction	point	none
M1 bias-corrected simulation	M0 plus the median offset	point	q_p
M1s scaled simulation	$a + b$ M0, least squares on the calibration alternatives	point	q_p
M2 envelope + residual margin	upper end of the envelope plus the median offset	largest residual	q_p
M5 GP correction of the envelope	envelope plus a Gaussian-process offset over the descriptors	GP, 90 %	q_p
M5n GP of q_p	Gaussian process of q_p over the descriptors	GP, 90 %	q_p
NN nearest produced setting	mean q_p at the nearest produced process setting of the family	point	q_p
M0w process-window worst case	lowest to highest simulated value over the incoming conditions	range	none
Mc friction-calibrated simulation	simulation with friction calibrated on the draw-in, plus the median offset	point	centres, q_p
M3 simulation + boosting	matched simulation plus a boosting model of the discrepancy, part level	jackknife+	parts
M3n boosting, no simulation	boosting model of the characteristic, part level	jackknife+	parts
M1n median of the produced	median q_p of the calibration alternatives	point	q_p
M2n median + residual margin	M1n with the largest-residual margin	largest residual	q_p

Descriptors: geometry indicator, blank-holder force and lubrication rank (the order of the oil amount). GP: noise variance bounded below by the variance between the series of one geometry and force. M3n is named M4 in the released code; M6, a conformalised GP, appears only in the robustness analysis.

trades decisiveness for safety, as the cost of errors would set the threshold of a reject option (Chow, 1970); 0.9 is used here, and 0.8 and 0.95 are reported in the robustness analysis.

3.4 Operating characteristics: decisive and safe distances

A rule is scored by judging each produced alternative as a new design at requirements placed at a known distance from the truth,

$$R = q_p(a, c) + d F_c, \quad (4)$$

so that a is adequate exactly when $d \geq 0$. For a case (a, c) with interval $[L, U]$, let $u = (U - q_p)/F_c$ and $v = (q_p - L)/F_c$ be the distances by which the interval excludes the truth from above and below. At $d = +\delta$ the verdict is correct when $u \leq \delta$ and a false reject when $-v > \delta$; at $d = -\delta$ it is correct when $v < \delta$ and a false accept when $-u \geq \delta$. Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and of wrong verdicts among all feasible verdicts at $|d| = \delta$: every case gives one verdict on an adequate alternative

and, where $q_p - \delta F_c \geq 0$, one on a failing alternative, and every verdict has equal weight. As δ grows, failing-side requirements become infeasible for characteristics close to zero, so the failing side carries half the weight up to a few floors and a third at about 100 floors (ESM Table S11). The *decisive distance* δ_d of a rule is the smallest δ beyond which $\kappa \geq 0.95$, and its *safe distance* δ_s the smallest δ beyond which $\omega \leq 0.05$. The distances are exact: κ and ω change only where δ crosses one of the case's u , v , $-u$ or $-v$, so the distances are order statistics of these exclusion distances; for a rule that always decides, $\delta_d = \delta_s$ is close to the 90 % quantile of its absolute error in floors. Always $\delta_s \leq \delta_d$, and between the two the rule sends designs to trial rather than misleading. The distances are reported for both sides together and one-sided, for failing alternatives (false accepts) and adequate ones (false rejects), which matters when the errors cost differently. They are points on operating characteristics, functions of the true margin over a population of design situations: the decisive distance is the analogue of the indifference zone of a selection procedure (Bechhofer, 1954) at a probability of correct selection of 0.95, and the pair plays the role of the acceptable and rejectable quality levels of an acceptance plan. A design team uses them before a requirement is fixed, to choose a rule for a family and evidence relation, to judge whether a requirement near the expected capability can be decided at the design stage at all, and to plan trials and production evidence. A trial is itself not error-free: a short run reproduces the 95th percentile only to about one floor (Section 5.1), so no decision can be relied on closer to the requirement than the trial that would replace it.

3.5 Evidence relation, calibration budget and applicability guard

How a new design relates to the produced alternatives is part of the decision situation. With a *sibling*, an alternative at the same process setting has been produced with another variation that production may not resolve; a *new process setting* lies between produced settings (interpolation) or outside them (extrapolation); and a *new family* shares no produced alternative. The distances are evaluated for each relation. Beyond a factorial, the relation can be graded by a distance in descriptor space or by the effect-to-scatter ratio between the predicted characteristic of the new design and its nearest produced alternative (Section 6.5). The calibration budget is the dependence of the distances on the number k of produced alternatives, over all subsets of size k classified by their relation to the new design. An applicability guard returns *trial* when the new design lies outside what the calibration has seen: its geometry is not among the calibration alternatives (family guard), a descriptor lies outside the calibrated range (range guard), or its simulated q_p lies outside the range of the calibration alternatives' simulated values (novelty guard), in the spirit of a valid input domain (Malak and Paredis, 2010).

3.6 Procedure

Figure 1 turns the framework into six steps: steps 1 to 5 once per design family, repeated when production evidence is added, and step 6 for every new design. Step 1 frames the decision (family, alternatives, characteristics, form of the requirement and

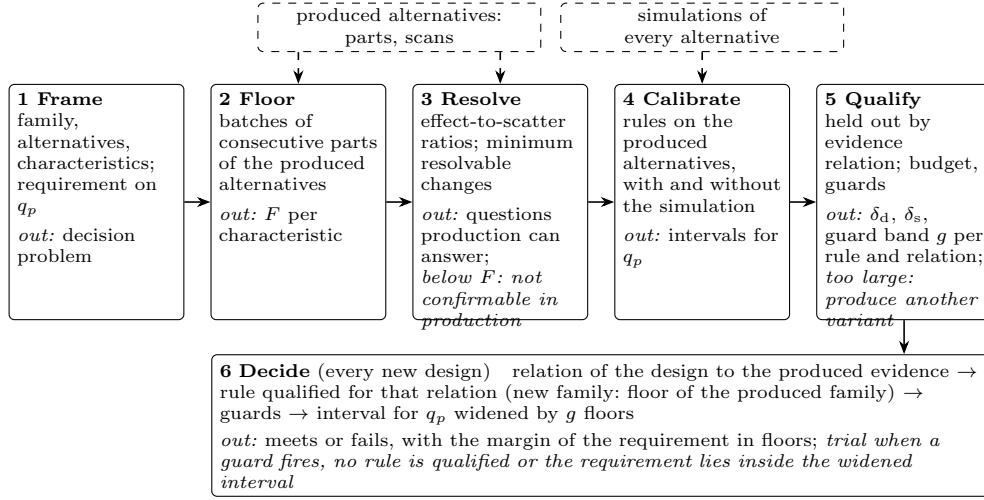


Fig. 1 The procedure: steps 1–5 once per design family, step 6 for every new design; dashed boxes are the evidence it uses

conformance level); step 2 measures the floor with a stated protocol; step 3 states which differences between alternatives, and which local changes of a parameter, production can resolve; and step 4 calibrates the candidate rules with and without the simulation. Step 5 qualifies them by their distances for each evidence relation and fixes for each rule and relation a *guard band* g : the smallest observable margin between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts were correct, for requirements within 20 floors of the truth. If the distances are too large for the requirements of coming designs, another variant is produced. Step 6 establishes the relation of the new design, applies the rule qualified for it and the guards, widens the rule’s interval by g floors and returns meets or fails with the margin in floors, or a trial when a guard fires, no rule is qualified or the requirement lies inside the widened interval.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzelmann, 2026; Baum et al, 2026a). Square blanks of DP600 dual-phase steel, 210 mm wide and about 1 mm thick, were deep drawn into cups (operation OP10) and cut (OP20) on one tool set with two geometries, concave and convex side walls, which differ in the curvature of the base shape, the wall angle and the bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns (coarse, medium and fine oil application, with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 g/m² of oil), giving 18 alternatives, each produced as one series of 500 consecutive parts; the change of geometry is the only change of product design. For every part the dataset holds

the press signals, traverses of the sheet thickness and of the oil film before forming, and 3D scans after drawing and after cutting. Of the 9,000 parts, 10 carry no scan. The series start at punch temperatures of 20.5–29.6 °C and warm up by 1.0–5.0 K over their first 150 parts; their median sheet thickness ranges from 0.986 to 0.993 mm, and the 100 kN series ran at a stroke speed of 47 mm/s against 78–79 mm/s for the others (ESM Table S10). The documentation states neither the order and dates of the series nor whether the parts were scanned in production order.

4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) of the same group (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025): LS-DYNA models of the same process on a quarter of the part, with nodes about 1 mm apart, a scaled DP600 flow curve, Coulomb friction with a constant coefficient and springback after both operations; the documentation does not state the yield criterion or the unloading model, which govern springback (Yoshida and Uemori, 2002), and the models do not represent the stroke speed. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm) and the friction coefficient (0.05–0.15) at each geometry and blank-holder force. A part is matched to the simulation with the nearest sheet thickness and the friction coefficient that the dataset’s linear mapping assigns to its oil film, an assumption of the dataset’s authors on which Mc, M3 and every matched simulation depend. The nominal simulation of an alternative is the one at 0.99 mm, the grid value nearest the measured median thickness, and friction 0.10, the middle of the range.

4.3 Quality characteristics and their measurement

Seven characteristics were extracted with the same definitions from the scans and from the simulated nodes, all referred to the plane of the cup bottom (ESM Table S1): the mid-side and corner draw-in and the flange waviness, process indicators of material flow and wrinkling on the flange that the cut removes, and four characteristics that a drawing would carry: the wall angle (design angle 20° concave and 10° convex), the cup depth, and the arm angle and bottom dome of the cut part, the last two with requirements on their absolute values. The four sides of a part measure several of them twice, and this redundancy exposes the measurement (ESM Table S8): the flange width along the scan lines scatters 4 to 18 times more than across them, so the mid-side draw-in is taken across the scan lines only, and the north wall carries uncorrelated scatter and the south wall is shadowed, so the wall angle is the mean of the east and west walls. The mid-side draw-in is quantised in steps of 0.019 mm, 0.38 floors, by the pixel pitch; the other characteristics are effectively continuous. Differences between scans and simulations, such as the scanned surface against the shell mid-surface, enter the absolute offsets between the two but not the floors.

4.4 Evaluation design

Every produced alternative is judged as a new design under grouped cross-validation (Roberts et al, 2017), one scheme per evidence relation. *New variant*: the alternative

alone is held out and the other eight alternatives of its geometry calibrate, so two siblings at its blank-holder force are among them. *New process setting*: the three alternatives of one force of a geometry are held out together and the six others calibrate; the 300 kN alternatives are interpolated and the 100 and 500 kN alternatives extrapolated in force. *New family*: the alternatives of one geometry calibrate the rules for the other, which gives two transfers. Pooled versions of the first two add the other geometry’s alternatives to the calibration set and test whether the production record of another family helps. Every calibrated rule is evaluated with and without the simulation (M1/M1n, M2/M2n, M3/M3n, M5/M5n), so that the paired difference isolates what the simulation contributes as it is used, and the nearest produced setting is the baseline without simulation and learning. A scope comprises 126 cases (18 alternatives, 7 characteristics).

The distances are computed exactly. Their intervals come from 1000 resamples of the held-out alternatives within each geometry, each with a block-bootstrap replicate of the true q_p of every alternative, common to all rules so that their differences get paired intervals; these intervals hold the fitted rules fixed. A second bootstrap resamples whole lubrication patterns within each geometry, which keeps every case in its evidence relation, and refits every alternative-level rule (ESM Section S1).

4.5 Computation and development record

The constants of the floor and the requirement grid were fixed before the decision results were seen, as the repository history records. The rules were not: M0–M5 were defined in a first analysis; M0w, Mc, M1n, NN, M2n, M5n, the grouped schemes, the exact estimator and the noise floor of the Gaussian processes were added after the first results had been seen; and a review of that version added M1s, the jackknife+ intervals, the between-run noise floor and the analyses of Sections 5.4 and 5.7. The comparison of rules is therefore exploratory and carries the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010); every variant tried is reported in the paper or the ESM. The pipeline reads single files from the public archives and reproduces every result from the cached feature tables in about four hours on four processor cores.

5 Results

5.1 What the production floor contains

In 95 % of batch pairs, two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in and 0.078° in wall angle (Table 3), and one floor is 0.66 to 2.2 long-term standard deviations of the parts. The floors are set by drift within the run: without the time order of the parts they shrink by a factor of 2.1 to 4.5, batches one position apart differ by 0.51–0.80 floors and nine positions apart by 1.1–1.5 floors, and the normal model of Section 3.2 reproduces the floors to within 12 % (ESM Tables S3 and S4). Independent part-to-part noise, which includes the repeatability of the scanner, would produce 15–33 % of each floor, and the resolution of the measurement is small against the floor except for the mid-side draw-in (ESM

Table 3 Production floor per characteristic, pooled and per geometry

Characteristic	F	95 % interval		Concave	Convex	F/σ_{LT}	Drift	Step
		batches	series					
Draw-in, mid-side (mm)	0.051	0.045–0.061	0.042–0.061	0.050	0.052	1.2/1.5	3.3	0.38
Draw-in, corner (mm)	0.046	0.035–0.052	0.032–0.053	0.053	0.030	1.7/1.3	4.5	<0.01
Flange waviness (mm)	0.0028	0.0024–0.0031	0.0021–0.0050	0.0037	0.0021	2.1/0.8	2.5	<0.01
Wall angle (°)	0.078	0.066–0.10	0.054–0.10	0.081	0.078	1.6/0.7	2.1	<0.01
Arm angle, cut (°)	0.11	0.094–0.12	0.089–0.15	0.12	0.100	1.9/0.9	2.6	<0.01
Cup depth (mm)	0.011	0.0096–0.012	0.0090–0.012	0.0087	0.012	2.2/1.7	3.9	<0.01
Bottom dome, cut (mm)	0.0070	0.0061–0.0081	0.0052–0.0084	0.0034	0.0082	1.4/1.5	3.9	<0.01

Batches of 50 consecutive parts, 95 % quantile of the difference of batch centres. Intervals: 2000 resamples of batches within series, and of whole series. F/σ_{LT} : floor over the median long-term standard deviation of the parts (concave/convex). Drift: floor over the floor of randomly permuted series. Step: measurement resolution in floors.

Table S2); assumption A3 holds for repeatability and resolution, but the stability of the scanner cannot be checked without a reference artefact. Nor do the process signals identify the source of the drift: batch centres of press force, punch temperature, sheet thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20 , and removing the predicted drift changes the floors by -16 to $+12$ % (ESM Table S30). Variation between runs is not in the floor: the three lubrication series of one geometry and force differ by up to 1.2 (waviness) to 9.8 floors (dome), and a floor taken across these series is 1.5 to 9.1 times the reference. The decided statistic reproduces to about one floor: the 95th percentiles of two batches of 100 parts differ by 0.92 – 1.49 floors in 95 % of pairs. Distances below about one floor are therefore beyond what production, or a trial, can confirm.

5.2 What production resolves and how far the simulation misses it

All 63 geometry contrasts exceed the floor and blank-holder force is resolved in 105 of 126 contrasts, but the lubrication pattern only in 51 of 126 (42 with a bootstrap probability of at least 0.95), so a lubrication variant is often a near-replicate of its siblings (ESM Table S5). The minimum resolvable force change is 21 – 67 kN between 100 and 300 kN for most characteristics and lies beyond either force interval for the wall angle and the dome. The nominal simulation misses production by far more than a floor: its median offset reaches 109 floors, its blank-holder force effect on the draw-in is 2.7 to 5.1 times the measured one, and the cut arms of the concave cups bend downwards in 89 – 100 % of the simulations but upwards in every part (ESM Tables S6 and S7). Part of the offset is definitional: the cup depth differs by a median 0.90 and 0.95 mm in the two geometries, consistent with the scanned surface against the shell mid-surface, and removing this offset alone brings the decisive distance of the nominal simulation from 108 to 65 floors.

5.3 Decisions by evidence relation

Table 4 and Fig. 2 score the rules. The nominal simulation needs requirements 108 floors from the truth (bootstrap interval 101–115) before 95 % of its verdicts are right; it errs mostly by rejecting adequate alternatives, with a safe distance of 17 floors for false accepts but 109 for false rejects.

A new variant with a produced sibling. With the other eight alternatives of the family produced, every calibrated rule improves on the nominal simulation, and the simplest is the most decisive: the nearest produced setting, the mean q_{95} of the two siblings at the same force, is right beyond 3.4 floors (2.0–4.5), 0.17 mm of mid-side draw-in. The scaled simulation follows at 4.9 floors, the Gaussian processes with and without the simulation decide at 7.2 floors and are safe below one floor, and the envelope rule is safe at 0.2 floors but sends a third of the designs to trial at 10 floors and decides only at 23 floors. Bias-correcting the nominal simulation (M1, 15 floors) is worse than taking the median of the produced alternatives without it (M1n, 9.1).

A new process setting. When every alternative at the new design’s force is held out, all rules lose. The nearest produced setting is right beyond 8.5 floors (5.5–10.2); the interval rules decide only at 13–31 floors and are safe at 5.9–16, and at 10 floors they send 18–37 % of the designs to trial. Interpolating in force (300 kN from 100 and 500 kN) is much easier than extrapolating: the nearest setting decides at 4.7 against 9.5 floors, and M2 and M5 are safe at 2.7 and 2.8 against 9.9 and 8.9 floors. The scaled simulation decides at 19 floors and, extrapolating a fitted trend, falsely accepts failing alternatives as far as 164 floors from their limit.

A new design family. In the two transfers, every rule without the simulation fails (131–133 floors), because nothing in one family’s production record describes the other. With the simulation, the bias correction decides at 35 floors, M5 at 38 and the boosting model M3 at 49, and the envelope rule is the safest at 16 floors; no rule is safe closer than that. Scored in the floor of the produced family, which a design team would have, M5 decides at 35 and is safe at 31 floors and the envelope rule at 41 and 26. Sixteen floors are about 0.8 mm of mid-side draw-in or 1.3° of wall angle.

5.4 Where the distances come from

Three features of the evaluation design shape the distances (ESM Table S21). First, the sibling relation mixes replication with variation. In 58 of the 126 new-variant cases neither sibling differs resolvably from the held-out alternative, and the nearest produced setting is then right beyond 0.85 floors, the reproducibility of q_{95} between near-replicate runs; when both siblings differ resolvably (34 cases), it needs 4.9 floors and M5 9.7. Second, the extrapolation cases are confounded with stroke speed: the production-only rules lose almost only at 100 kN, where the press also ran slower (the nearest setting decides at 11 floors against 4.9 at 500 kN), whereas the simulation-based interval rules are unsafe at both ends. Third, the production record of the other family does not help within a family: pooling it makes most rules less decisive (M5 from 7.2 to 11 floors for a new variant, M2 from 23 to 53, M2n from 19 to 325), although it raises the coverage of M5 from 86 to 90 % and makes the centre of its

Table 4 Decisive and safe distances (floors) by the relation of the new design to the produced evidence

Rule	New variant		New process setting				New family	
	δ_d/δ_s	FA	δ_d/δ_s	FA	interpolated	extrapolated	δ_d/δ_s	FA
M0	108	17	108	17	109	106	108	17
M1	15	15	24	24	11	25	35	33
M1s	4.9	4.7	19	164	19	19	122	122
interval	4.7–6.3		14–21		19–20	15–19	121–123	
M2	23/0.2	0.3	18/8.1	8.3	18/2.7	18/9.9	51/16	11
M5	7.2/0.8	0.2	19/6.8	7.6	19/2.8	19/8.9	38/28	32
interval	6.6–8.4		18–20		18–20	18–20	35–39	
M5n	7.2/0.6	0.1	17/6.1	9.0	16/0.0	18/8.6	132/130	167
interval	6.4–7.5		16–18		15–16	16–19	130–133	
NN	3.4	2.8	8.5	10	4.7	9.5	131	169
interval	2.0–4.5		5.5–10		4.3–5.1	6.6–11	129–131	

Over all seven characteristics; one number for rules that always decide. FA: safe distance on the false-accept side (failing alternatives). Interval: bootstrap 95 % interval of the decisive distance (fits fixed). 126 cases per relation (42 interpolated, 84 extrapolated); a new family rests on two transfers. All rules: ESM Table S12.

interval the most accurate point prediction within the family (2.9 floors against 3.4 for the nearest setting).

The relation therefore does not govern decision fitness in general. Over the eleven calibrated rules and all three relations, the relation explains 69 % of the variation of the log decisive distance and the rule 6 %, but this share comes from the new family, where the rules without the simulation fail by construction; within a family the rule explains 68 % and the relation 15 %. What the data show is that the best attainable distance grows with the novelty of the design, from 3.4 floors with a sibling to 8.5 for a new setting and to 35 for a new family, and that within a family the choice of rule matters at least as much.

5.5 What the simulation and learning contribute

Pairing each calibrated rule with its counterpart without the simulation isolates what the simulation adds as it is used (ESM Table S15). Added as an offset, it adds nothing within a family and degrades two rules: the decisive distance of M5 equals that of M5n (difference 0.0 floors, -0.4 to $+1.1$), and those of M2 and M1 exceed those of M2n and M1n by 3.9 ($+0.5$ to $+7.4$) and 6.2 floors ($+4.6$ to $+9.7$), because the simulated force effect is too large. Rescaled, it adds information within a family, where the scaled simulation decides 4.2 floors closer than the median of the produced alternatives (-5.4 to -0.5) though 1.5 floors less close than the nearest produced setting, but for a new setting the rescaled trend is worse than none. Only for a new family is the simulation decisive, by about 94 floors, and even there production knowledge carries over where a drawing nominal transfers: for the wall angle, the design angle plus the produced family's deviation from its own design angle decides at 2.0 floors, against 7.3 for M5 (ESM Table S19). A surrogate of the simulations reproduces them to within 1.8 floors

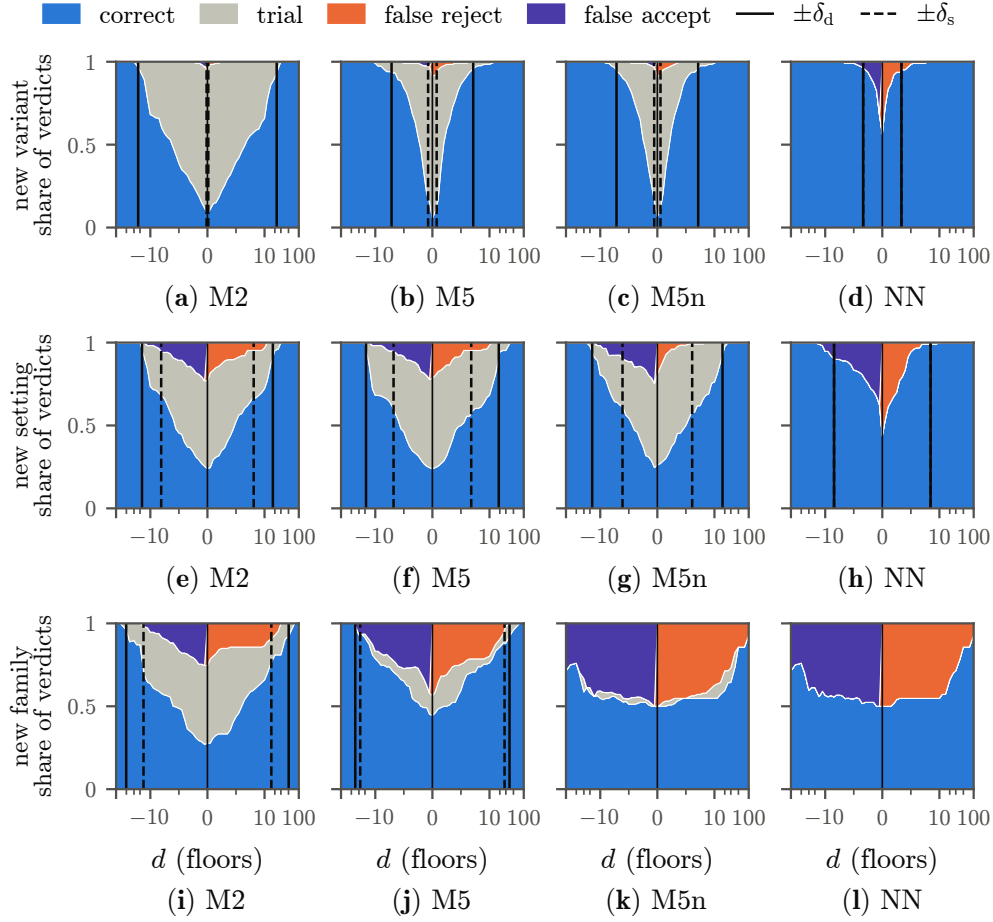


Fig. 2 Verdict shares against the requirement distance d (symmetric logarithmic axis, linear within ± 10 floors; 126 cases per panel) with the decisive and safe distances marked

for 12 of the 14 characteristics and geometries, yet its decision fitness is that of the simulation it reproduces.

Learning adds intervals rather than accuracy. Used as point rules, the centres of the M5, M5n and M3n intervals decide at 3.3, 3.1 and 3.7 floors within the family, as close as the nearest produced setting, so the ranking of the learned rules rests on the width of their intervals (ESM Table S16). With the noise floor set by the variation between runs, the Gaussian-process intervals cover 85–86 % of the true q_{95} within the family against a nominal 90 %, the largest-residual margin 89 % (it guarantees at least $(n-1)/(n+1) = 7/9$ under exchangeability, which these cases lack) and the jackknife+ interval of M3 96 %; for a new setting every calibrated interval covers only 43–55 %, and for a new family the Gaussian processes cover at most 12 %. The inductive biases explain much of this: trees do not extrapolate, two force levels identify at most a linear

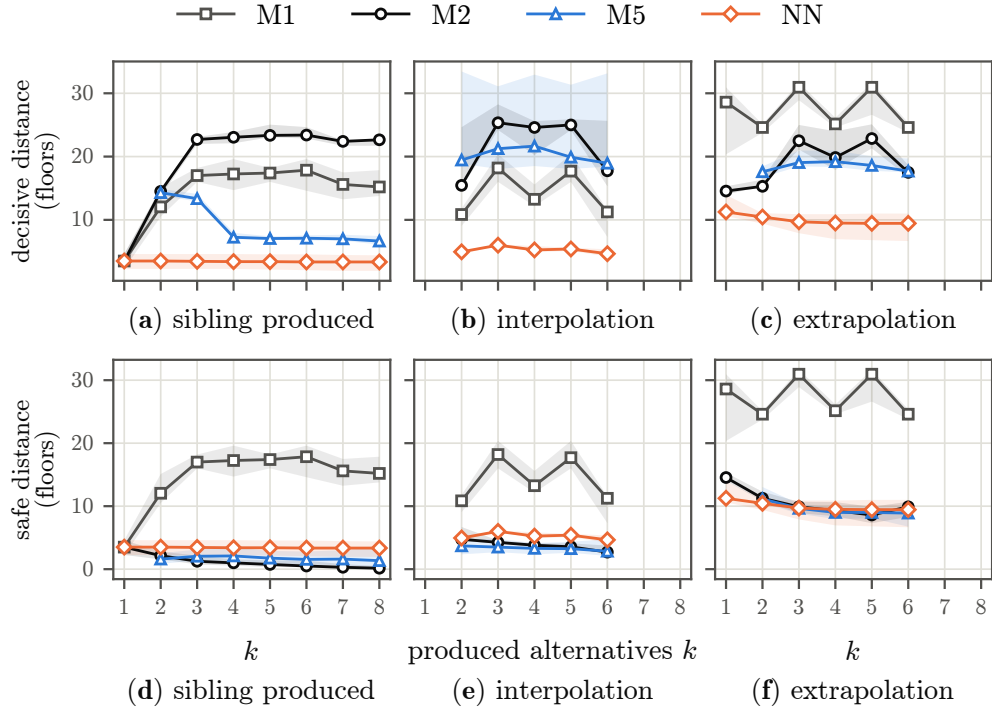


Fig. 3 Decisive and safe distances against the number of produced alternatives of the family, by the relation of the new design to them; bands: 95 % intervals over the held-out alternatives

trend, and in the median new-setting fit the length scale of the force sits at its lower bound, so the Gaussian process reverts to its mean (ESM Tables S23 and S22).

5.6 Calibration budget

Figure 3 and ESM Table S20 show how the distances depend on the number of produced alternatives of the family and on how the new design relates to them. A single produced sibling decides: with one alternative at the new design's force, the bias correction, the envelope rule and the nearest setting coincide by construction (the simulation terms cancel and a single residual gives no margin) and decide at 3.5 floors (2.3–4.6) without a wrong verdict at 10 floors. Further alternatives do not improve the nearest setting; they make the bias correction worse (12–18 floors), because the offsets of other forces dilute the sibling's, and the interval rules safer but not more decisive. Without a sibling, bracketing the new force buys safety but not decisiveness: *[budget numbers pending]*. Extrapolation stays unsafe at every budget, and averaged over all subsets the distances shrink with the number of produced alternatives mainly because the share of subsets that contain a sibling grows, from 46 % with two alternatives to 96 % with six. With calibration series of 50, 100 or 250 parts instead of 500, *[short-series numbers pending]*.

5.7 The procedure's decision rule

What a designer observes for a single verdict is the margin between the predicted interval and the requirement, and its value as evidence depends on the relation (Fig. 4): for a new variant at least 95 % of the decided verdicts of the interval rules are correct at every margin; for a new setting the share grows with the margin, for the nearest setting from 85 % at any margin to 95 % beyond 5 floors; for a new family M5 is right in 71–74 % of its decided verdicts whatever the margin. Step 5 turns this into guard bands (ESM Table S24): none for the interval rules within the family and 0.25 floors for the nearest setting; 2.0 floors for an interpolated setting, with which the nearest setting decides at 6.7 floors, is safe at 2.7 and sends no design to trial at 10 floors; 7.0 floors for an extrapolated one, with which it decides at 17 floors and sends a quarter of the designs to trial at 10 floors; and for a new family no guard band up to 20 floors reaches 95 %, so the procedure sends every design to trial. The guard bands are chosen on the evaluation data, which makes the trial rates optimistic.

The applicability guards act on the same relations (ESM Table S25): for a new setting the range guard refuses exactly the extrapolated forces, after which M2, M5 and the nearest setting make no wrong verdict at 10 floors, whereas the novelty guard also refuses many cases that would have been decided correctly.

A worked example. A tryout engineer has to decide whether the concave cup at 300 kN with medium lubrication, a new process setting, can be released without a trial run, with the six alternatives at 100 and 500 kN produced and a requirement that 95 % of the parts have a mid-side draw-in of at most 15.30 mm. The floor is 0.050 mm, the relation is interpolation and no guard fires. The nominal simulation predicts 15.66 mm and fails the design. The nearest produced setting predicts 14.99 mm, a margin of 6.2 floors, more than its guard band of 2.0 floors, so step 6 returns meets; beyond a margin of 6 floors every decided verdict of this rule for an interpolated setting was correct. M2, M5 and M5n return intervals reaching 15.37, 15.58 and 15.33 mm and send the design to trial. The 500 parts had a 95th percentile of 14.88 mm, 8.4 floors below the limit: the procedure was right, the simulation wrong and the interval rules cautious.

5.8 Requirements anchored in capability and in tolerances

Upper limits at which an alternative has a performance index of 1.0, 1.33, 1.67 or 2.0 lie a median 1.0, 1.7, 2.5 and 3.1 floors above its 95th percentile, inside every decisive distance (ESM Table S26). Every alternative meets them, so a verdict is right, a false reject or a trial. At a performance index of 1.33 the nearest produced setting still says meets in 91 % of the new-variant cases, because most of its errors are far smaller than its 95 % distance, whereas M5 says so in 44 % and sends the rest to trial; for a new setting the two are right in 65 and 32 %, and for a new family no rule exceeds 54 %. Release-level requirements are therefore decided at the design stage only within a family, and then by the production record. General angular tolerances of ISO 2768-1 (ISO, 1989) give a coarser scenario that mostly tests whether a rule recognises gross failures: only 32 of its 108 decisions lie within 5 floors of the limit (ESM Table S27). Its cheapest rule depends on the costs (ESM Table S28): for a new variant the nearest setting at any cost, for a new setting the production-only interval rules while a trial

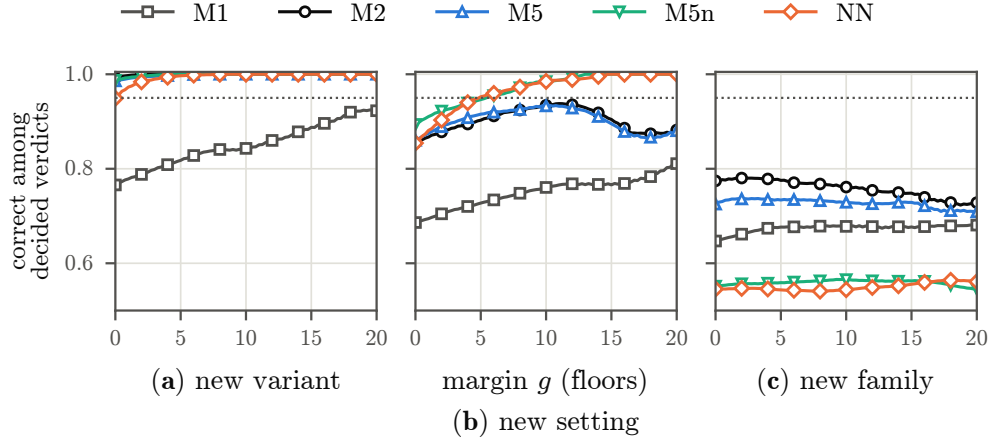


Fig. 4 Share of correct verdicts among the decided ones against the observable margin between interval and requirement, for requirements within 20 floors of the truth

costs at most a tenth of a false accept and the nearest setting beyond, and for a new family a trial of every design while a trial costs at most a tenth of a false accept, the envelope rule at a fifth and M5 beyond, for a false reject at half the cost of a false accept.

5.9 Robustness

The analysis was repeated under variants of the data, the constants and the model specifications (ESM Table S29). *[robustness summary pending]*

6 Discussion

6.1 What the evaluation establishes

Five findings answer the research questions. Two follow from the definitions and hold wherever the framework is applied. First (RQ1), the floor puts every rule and characteristic on a common production-referenced scale, but not on a common scale of nonconforming fractions, and as a within-run quantity it is a lower bound of the variation of series production. Second, an interval rule trades decisiveness for safety, the two distances make the trade explicit, and the nominal level of an interval is reached only where the calibration alternatives are exchangeable with the new design. Three are observations on this dataset and hypotheses for confirmation. Third (RQ2), the best attainable distance grows with the novelty of the design, from 3.4 floors with a sibling (0.85 when the sibling is a near-replicate) to 8.5 for a new setting (4.7 interpolated, 9.5 extrapolated with a speed change) and to no rule safe closer than 16 floors in two transfers, while within a family the choice of rule matters more than the relation. Fourth, production evidence buys safety before decisiveness: interval rules become safe with few produced alternatives that bracket the new design, and decisive only with near-replicates. Fifth (RQ3), in this small-data regime and with this simulator, the

Table 5 Guidance for design-stage manufacturability decisions, with its basis

Situation	Guidance	Basis
<i>From the framework (general)</i>		
Before trusting a predictor	Measure the floor on consecutive production of the family with a stated protocol and report σ_b , σ_w and its lag dependence; treat it as a lower bound of series variation.	definitions
Qualifying a rule	Evaluate it for the evidence relation of the coming designs by grouped cross-validation, and split near-replicate siblings from resolvable ones.	definitions
Choosing a rule	Compare it with the nearest produced setting and with its own version without the simulation, on decisive and safe distance jointly or by expected cost.	definitions
Deciding a single design	Apply the rule’s guard band to the observable margin; a margin inside the guard band, or a fired guard, means a trial.	definitions
Fixing a requirement	If the expected margin to the capability is below the safe distance of the best qualified rule, plan a trial; no decision is reliable closer than a trial reproduces the quantile.	definitions
<i>Observed on this dataset (18 alternatives, 2 geometries, 7 characteristics)</i>		
New variant with a produced sibling	nearest produced setting right beyond 3.4 floors (0.85 for a near-replicate); calibrated interval rules safe below one floor	126 cases
New process setting	nearest produced setting right beyond 4.7 floors interpolated and 9.5 extrapolated; interval rules safe at 5.9–16 floors	42 + 84 cases
New family	best rule safe at 16 floors; rules without the simulation fail; a transferable drawing nominal helps	two transfers
Release-level requirements	lie within 1–3 floors of the 95th percentile, inside every decisive distance	126 cases

simulation contributed to decisions only across families as an offset, a fitted scale made it useful within a family but harmful for a new setting, and learned models were as accurate as the nearest produced setting as point predictors while adding abstention.

6.2 Using the framework in design

Table 5 separates guidance that follows from the framework from observations on this dataset, with their basis. Before a requirement is fixed, the expected margin between requirement and capability can be compared with the safe distance of the best rule qualified for the relation of the design; if the margin is smaller, no design-stage decision can be relied on, and a trial or further production evidence has to be planned. Release-level requirements make this the usual case: they lie within a few floors of the capability, where only a near-replicate production record decides reliably, which is consistent with the practice of requiring a production trial run for a process change. For a single design, step 6 turns the qualification into a verdict whose guard band a designer can see, and the worked example shows that the most decisive rule is also the most transparent: the nearest produced setting points to the produced alternative

that it copies, whereas a Gaussian-process interval does not. Which rule to prefer then depends on the costs, and those differ by relation: a trial of a new force on an existing tool is a short press run, whereas a trial of a new family may need a new tool. In a company, steps 1 to 5 fit the start of production or the production part approval of a family, when its series are measured anyway, and step 6 the later decisions on variants and process changes.

973

974 **6.3 Simulation and machine learning in this regime**

975

976 Within a design family the production record carried the decision: the nearest pro-
977 duced setting was the most decisive rule, adding the simulation as an offset did not
978 improve a calibrated rule, and the bias correction made it worse, because the simulated
979 force trend was 2.7 to 5.1 times too large. Rescaling the simulation repaired this within
980 the family but extrapolated badly for a new setting, and calibrating the friction coef-
981 ficient did not repair it, which points to the friction law and the blank-holder model
982 rather than to a parameter. Learned models predicted as well as the nearest produced
983 setting; what they added was an interval that is safe where the calibration alterna-
984 tives are exchangeable with the new design and over-confident where they are not. A
985 multi-task model cannot help a family with no produced alternative, because the cor-
986 relation between families cannot be estimated without its data; the pooled Gaussian
987 process, which carries the geometry as an input, is the borrowing of strength that
988 the data allow, and it improved coverage but not decisiveness. These findings concern
989 one simulator of low fidelity for this production, used through offsets and one scale,
990 and learners with six to nine produced settings each; a surrogate of these simulations
991 inherits the simulator’s distance, whatever its accuracy against the simulation. For
992 the evaluation of AI-based DFM support the lesson is methodological: held-out accu-
993 racy is not enough, and a learned model should be compared with a production-record
994 baseline, scored by evidence relation in a unit that production defines, and shown to
995 its user with a guard band. Whether that makes designers’ reliance match a rule’s
996 reliability (Lee and See, 2004), and when an abstention should go to an expert rather
997 than to a trial (Madras et al, 2018), are hypotheses for a study with design teams.

998

999 **6.4 Implications for design research**

1000

1001 The framework gives several constructs of design research a production-referenced
1002 measure. For variant, adaptive and platform design (Pahl et al, 2007; Simpson et al,
1003 2006), the distances by evidence relation quantify how far production knowledge carries
1004 from produced variants to a new one, and the nearest produced setting sets the baseline
1005 that any capability database (Thornton, 2004) or learned model has to beat. For
1006 margins, the safe distance is the part of a margin that covers predictive uncertainty,
1007 as distinct from the part that absorbs change (Eckert and Isaksson, 2017; Eckert et al,
1008 2019), and the guard band is its counterpart for a single decision. For testing and
1009 prototyping strategies (Thomke and Bell, 2001; Salado and Kannan, 2018; Camburn
1010 et al, 2017), the distances and the reproducibility of a trial say when a test is cheaper
1011 than a decision and how close to a requirement a test can itself decide. For overlapped
1012 development (Krishnan et al, 1997; Terwiesch et al, 2002) and set-based design (Sobek

et al, 1999), they state when design-stage information is fit to act on and how wide a feasibility limit has to be. And for verification and validation, the evidence relation is the question of the application domain (Oberkampff and Roy, 2010), and the distances could serve as the evidence of a risk-informed credibility assessment (ASME, 2018). In the terms of the design research methodology (Blessing and Chakrabarti, 2009), the evaluation reported here is a support evaluation: it shows structural validity in the sense of the validation square, that the metrics discriminate between rules and relations on real data (Seepersad et al, 2006), and is a first step towards performance validity, which would require showing that decisions taken with the framework are better than decisions taken without it.

6.5 Reuse for other processes and AI-based DFM tools

The framework needs several produced variants of a design family whose relations to each other span those of the coming designs, a series of consecutive parts or replicated batches of one design per variant, a design-stage predictor for every variant, and characteristics measured identically on parts and predictions. Table A2 maps these onto the processes named most often in DFM; attribute outcomes such as splits, porosity or short shots call for a requirement on a rate rather than on a quantile, which the framework does not yet cover. A predictor need not return an interval: any tool that returns meets, fails or trial for a stated requirement, a classifier at its operating threshold or a language-model check given the requirement value, can be scored by sweeping the requirement, and an interval is sufficient but not necessary. For continuous design spaces, the relation becomes a distance from the produced designs in descriptor space, or the position inside or outside their hull. A minimal report of such an evaluation comprises the floor with its protocol and variance components, the decisive and safe distances by relation with refitting intervals and one-sided values, the coverage of interval rules, the nearest-produced-setting and no-simulation baselines, and the guard bands and guard performance. An attempt to apply the framework to an open dataset of powder bed fusion of PA12 (Leirimo, 2021) failed on the second requirement: grouping specimens by orientation class mixed orientations in a batch, and eight identical anchor specimens per build were too few for a floor; a between-build floor from replicated anchors would be the remedy.

6.6 Threats to validity

The evaluation covers one material, one press and two geometries, and the relations within a family come from a 3×3 factorial in which one factor, lubrication, is mostly not resolved; the procedure is general, the numbers are not. The new family rests on two transfers between geometries that differ in three features at once, and the intervals of its distances hold the two fits fixed. Each alternative was produced once, so the floor covers drift within a run but not between days, coils or tool states, the reference q_{95} is that of one run, and the order and dates of the series are unknown; between-series variation bounds the missing component at 1.2–9.8 floors, and a floor that includes it rescales the distances accordingly. A drift of the scanner cannot be separated from that of the process, so the floor is a production-and-measurement floor, and scan and reference-surface differences enter the absolute offsets of the simulation. The

1059 extrapolation in force is confounded with stroke speed, which the simulation does not
1060 represent, so the extrapolation results describe a force and speed change together. The
1061 requirements are one-sided limits on the 95th percentile of a characteristic averaged
1062 over sides, which is below the conformance of series release. Several rules, the grouped
1063 schemes and the guard bands were added or chosen after earlier results had been seen
1064 and are evaluated on the same data, so the comparison between rules carries selection
1065 optimism (Cawley and Talbot, 2010); the analysis was carried out by a single analyst,
1066 with code and data open for replication.

1067

1068 7 Conclusions

1069

1070 Design-stage manufacturability decisions can be evaluated against production on a
1071 scale that production defines. The production floor expresses the fitness of any pre-
1072 dictor for a decision in the units of the characteristic; the decisive and safe distances,
1073 read as operating characteristics, tell how close to a requirement a rule decides and
1074 how close it can be relied on not to mislead; and a guard band turns them into a rule
1075 for a single decision.

1076 On 18 deep-drawn design-process alternatives with matched simulations, the best
1077 attainable distance grew with the novelty of the new design: with a produced sibling
1078 the production record decided within 3.4 floors, for a new process setting no rule
1079 decided closer than 8.5 floors, and in two transfers to a new family no rule was safe
1080 closer than 16 floors. Within a family the choice of rule mattered more than the
1081 relation, release-level requirements lay inside every decisive distance, and in this small-
1082 data regime the simulation as used, a model with a large and partly sign-inconsistent
1083 bias that does not represent the stroke speed, contributed only across families, while
1084 learned models added abstention rather than accuracy.

1085 Four extensions follow: series produced on several days, coils and tool states, so
1086 that the floor covers variation between runs; design families with continuous geometric
1087 descriptors; a confirmatory evaluation of the frozen procedure on another dataset or
1088 process; and an evaluation with design teams of whether decisions taken with the
1089 framework are better. The released code, which reproduces every result from the public
1090 data, is the starting point for all four.

1091

1092 **Supplementary information.** The Electronic Supplementary Material contains
1093 Tables S1–S30 and Fig. S1 with the further results referred to in the text.

1094 **Acknowledgements.** The author thanks the authors of the RDDAC and DDACS
1095 datasets at the University of Stuttgart for publishing the data on which this work rests.

1096

1097 Declarations

1098

1099 **Funding.** No specific funding was received for this work.

1100

1101 **Competing interests.** The author declares no competing interests.

1102 **Ethics approval and consent.** Not applicable.

1103

1104

Data availability. All data are public. RDDAC: doi:10.18419/DARUS-5589 (version 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0. The extracted per-part and per-simulation feature tables are provided with the code.

Code availability. The complete analysis pipeline is released under an open licence at <https://github.com/htcanseven/Claude> (directory `scripts/`) and reproduces every number in this paper from the public data; a versioned archive of the code and the feature tables is deposited on Zenodo [*DOI pending*].

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author’s direction as an assistant for code development, literature retrieval, data analysis, drafting and editing of the text, and for simulated peer review of earlier versions. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. No AI-generated images are used; all figures are produced by the released code from the data.

Appendix A Notation and reuse

Table A1 summarises the notation and the terms defined in Section 3, and Table A2 maps the framework’s requirements onto other processes. Computation details are given in the ESM.

References

- Ahmed F, Picard C, Chen W, et al (2025) Special issue: Design by data: Cultivating datasets for engineering design. *Journal of Mechanical Design* 147(4):040301. <https://doi.org/10.1115/1.4067871>
- Angelopoulos AN, Bates S, Fisch A, et al (2024) Conformal risk control. In: *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7–11, 2024*. OpenReview.net, URL <https://openreview.net/forum?id=33XGfHLtZg>
- ASME (2018) *Assessing credibility of computational modeling through verification and validation: Application to medical devices*. ASME V&V 40-2018, American Society of Mechanical Engineers, New York
- Automotive Industry Action Group (2006) *Production Part Approval Process (PPAP)*, 4th edn. AIAG, Southfield, MI
- Automotive Industry Action Group (2010) *Measurement Systems Analysis*, 4th edn. AIAG, Southfield, MI

Table A1 Notation and terms

Symbol, term	Meaning
$a, c, q_p(a, c)$	alternative (design–process combination), quality characteristic, p -quantile of the parts of a ($p = 0.95$)
R, d	requirement (upper limit on c or $ c $); its distance from the truth in floors, $R = q_p + dF_c$
$[L_r, U_r], V_r$	interval of rule r for q_p ; verdict meets, fails or trial (Eq. 1)
B, m, F_c	batch size (50); batch centre (10 % trimmed mean); production floor (Eq. 2)
σ_b, σ_w	standard deviation between batch means and between parts within a batch
ESR, MRC	effect-to-scatter ratio (Eq. 3); minimum resolvable change, $F_c/ s_c $
u, v	exclusion distances of an interval from the truth, from above and below, in floors
$\kappa(\delta), \omega(\delta)$	shares of correct and wrong verdicts among the feasible verdicts at $ d = \delta$
δ_d, δ_s	decisive distance ($\kappa \geq 0.95$ beyond it) and safe distance ($\omega \leq 0.05$ beyond it)
FA, FR side	one-sided distances for failing alternatives (false accepts) and adequate ones (false rejects)
g	guard band: observable margin beyond which $\geq 95\%$ of decided verdicts were correct
Calibration alternatives	produced alternatives whose record calibrates a rule
Evidence relation	sibling (same process setting produced), new process setting (interpolation, extrapolation), new family
Calibration budget	distances against the number k of produced alternatives, by relation
Guard	family, range and novelty checks that send a design outside the calibration to trial

Table A2 What the framework requires, mapped onto other processes

Process	Batch and run	Dominant variation	Evidence relations	Floor
Sheet forming (this paper)	consecutive parts of one series	drift within a run, coil and die state between runs	variant, process setting, family	batch centres within a run
Injection moulding	consecutive shots of one cavity; one mould set-up	cavity, shot, set-up	cavity, process window, new mould	nested: within cavity, between cavities
Machining	consecutive parts between offsets	tool wear, compensated by offsets	tool path, cutting data, new part	between offset intervals
Casting	castings of one pattern and melt replicated	melt, pattern position	alloy, gating, new pattern	between melts
Additive manufacturing	specimens of one build	build, position, orientation	orientation, parameters, new geometry	between builds, from replicated anchors

Ballegeer M, Benoit DF (2026) BenDFM: A taxonomy and synthetic CAD dataset for manufacturability assessment in sheet metal bending. Journal of Intelligent Manufacturing <https://doi.org/10.1007/s10845-026-02857-9>

Banabic D (2010) Sheet Metal Forming Processes: Constitutive Modelling and Numerical Simulation. Springer, Berlin, Heidelberg, https://doi.org/10.1007/978-3-540-88113-1	1197 1198 1199 1200
Barber RF, Candès EJ, Ramdas A, et al (2021) Predictive inference with the jack-knife+. The Annals of Statistics 49(1):486–507. https://doi.org/10.1214/20-aos1965	1201 1202 1203
Baum S, Heinzelmann P (2025) Deep drawing and cutting simulations (DDACS) dataset. DaRUS, University of Stuttgart, version 3.0, https://doi.org/10.18419/DARUS-4801	1204 1205 1206
Baum S, Heinzelmann P (2026) Real deep drawing and cutting (RDDAC) dataset. DaRUS, University of Stuttgart, version 2.0, https://doi.org/10.18419/DARUS-5589	1207 1208 1209 1210
Baum S, Heinzelmann P, Clauß P, et al (2026a) Statistical analysis of simulation-to-reality deviation in deep drawing with a benchmark dataset. Transactions of the Indian Institute of Metals 79(7):176. https://doi.org/10.1007/s12666-026-03870-5	1211 1212 1213 1214
Baum S, Heinzelmann P, Liewald M, et al (2026b) Addressing discrepancies between synthetic and real-world data of deep-drawing manufacturing systems in AI applications. Procedia CIRP 138:727–732. https://doi.org/10.1016/j.procir.2026.01.125	1215 1216 1217 1218
Bechhofer RE (1954) A single-sample multiple decision procedure for ranking means of normal populations with known variances. The Annals of Mathematical Statistics 25(1):16–39. https://doi.org/10.1214/aoms/1177728845	1219 1220 1221 1222
Ben Slama M, Chatti S, Louhichi B (2024) Novel method for shape complexity evaluation: a threshold from machining to additive manufacturing in the early design phase. Research in Engineering Design 35(2):191–214. https://doi.org/10.1007/s00163-023-00429-z	1223 1224 1225 1226
Blessing LTM, Chakrabarti A (2009) DRM, a Design Research Methodology. Springer, London, https://doi.org/10.1007/978-1-84882-587-1	1227 1228 1229
Boothroyd G, Dewhurst P, Knight WA (2010) Product Design for Manufacture and Assembly, 3rd edn. CRC Press, Boca Raton, https://doi.org/10.1201/9781420089288	1230 1231 1232 1233
Brahma A, Wynn DC (2020) Margin value method for engineering design improvement. Research in Engineering Design 31(3):353–381. https://doi.org/10.1007/s00163-020-00335-8	1234 1235 1236 1237
Brynjarsdóttir J, O’Hagan A (2014) Learning about physical parameters: the importance of model discrepancy. Inverse Problems 30(11):114007. https://doi.org/10.1088/0266-5611/30/11/114007	1238 1239 1240 1241 1242

1243 Camburn B, Viswanathan V, Linsey J, et al (2017) Design prototyping methods: state
 1244 of the art in strategies, techniques, and guidelines. *Design Science* 3:e13. <https://doi.org/10.1017/dsj.2017.10>
 1245
 1246
 1247 Cawley GC, Talbot NLC (2010) On over-fitting in model selection and subsequent
 1248 selection bias in performance evaluation. *Journal of Machine Learning Research*
 1249 11:2079–2107. URL <https://jmlr.org/papers/v11/cawley10a.html>
 1250
 1251 Chase KW, Parkinson AR (1991) A survey of research in the application of tolerance
 1252 analysis to the design of mechanical assemblies. *Research in Engineering Design*
 1253 3(1):23–37. <https://doi.org/10.1007/bf01580066>
 1254
 1255 Chow CK (1970) On optimum recognition error and reject tradeoff. *IEEE Transactions*
 1256 *on Information Theory* 16(1):41–46. <https://doi.org/10.1109/TIT.1970.1054406>
 1257
 1258 Eaton CE, Campo KX, Longhurst A, et al (2026) Omitted technical measures: estab-
 1259 lishing evidence for the necessity of problem formulation and validation in system
 1260 alternative selection. *Research in Engineering Design* 37(3):24. <https://doi.org/10.1007/s00163-026-00488-y>
 1261
 1262 Eckert C, Isaksson O (2017) Safety margins and design margins: A differentiation
 1263 between interconnected concepts. *Procedia CIRP* 60:267–272. <https://doi.org/10.1016/j.procir.2017.03.140>
 1264
 1265
 1266 Eckert C, Isaksson O, Earl C (2019) Design margins: a hidden issue in industry. *Design*
 1267 *Science* 5:e9. <https://doi.org/10.1017/dsj.2019.7>
 1268
 1269 Friedman JH (2001) Greedy function approximation: A gradient boosting machine.
 1270 *The Annals of Statistics* 29(5):1189–1232. <https://doi.org/10.1214/aos/1013203451>
 1271
 1272 Geifman Y, El-Yaniv R (2017) Selective classification for deep neural networks. In:
 1273 *Advances in Neural Information Processing Systems*, vol 30. Curran Associates,
 1274 Inc., pp 4878–4887
 1275
 1276 Ghadai S, Balu A, Sarkar S, et al (2018) Learning localized features in 3D CAD models
 1277 for manufacturability analysis of drilled holes. *Computer Aided Geometric Design*
 1278 62:263–275. <https://doi.org/10.1016/j.cagd.2018.03.024>
 1279
 1280 Hazelrigg GA (1998) A framework for decision-based engineering design. *Journal of*
 1281 *Mechanical Design* 120(4):653–658. <https://doi.org/10.1115/1.2829328>
 1282
 1283 Heinzelmann P, Baum S, Riedmüller KR, et al (2025) A comprehensive bench-
 1284 mark dataset for sheet metal forming: Advancing machine learning and surrogate
 1285 modelling in process simulations. *MATEC Web of Conferences* 408:01090. <https://doi.org/10.1051/mateconf/202540801090>
 1286
 1287
 1288

Hol J, Cid Alfaro MV, de Rooij MB, et al (2012) Advanced friction modeling for sheet metal forming. <i>Wear</i> 286–287:66–78. https://doi.org/10.1016/j.wear.2011.04.004	1289 1290 1291
ISO (1989) General tolerances – part 1: Tolerances for linear and angular dimensions without individual tolerance indications. ISO 2768-1:1989, International Organization for Standardization, Geneva	1292 1293 1294 1295
ISO (1994) Accuracy (trueness and precision) of measurement methods and results – part 6: Use in practice of accuracy values. ISO 5725-6:1994, International Organization for Standardization, Geneva	1296 1297 1298 1299
ISO (2017) Geometrical product specifications (GPS) – inspection by measurement of workpieces and measuring equipment – part 1: Decision rules for verifying conformity or nonconformity with specifications. ISO 14253-1:2017, International Organization for Standardization, Geneva	1300 1301 1302 1303
ISO (2025) Accuracy (trueness and precision) of measurement methods and results – part 2: Basic method for the determination of repeatability and reproducibility of a standard measurement method. ISO 5725-2:2025, International Organization for Standardization, Geneva	1304 1305 1306 1307 1308
ISO (2026) Statistical methods in process management – capability and performance – part 2: Process capability and performance of time-dependent process models. ISO 22514-2:2026, International Organization for Standardization, Geneva	1309 1310 1311 1312
Jiao JR, Simpson TW, Siddique Z (2007) Product family design and platform-based product development: a state-of-the-art review. <i>Journal of Intelligent Manufacturing</i> 18(1):5–29. https://doi.org/10.1007/s10845-007-0003-2	1313 1314 1315 1316
Joint Committee for Guides in Metrology (2012) Evaluation of measurement data – the role of measurement uncertainty in conformity assessment. JCGM 106:2012, BIPM, Sèvres, https://doi.org/10.59161/jcgm106-2012	1317 1318 1319 1320
Kennedy MC, O’Hagan A (2000) Predicting the output from a complex computer code when fast approximations are available. <i>Biometrika</i> 87(1):1–13. https://doi.org/10.1093/biomet/87.1.1	1321 1322 1323 1324
Kennedy MC, O’Hagan A (2001) Bayesian calibration of computer models. <i>Journal of the Royal Statistical Society Series B: Statistical Methodology</i> 63(3):425–464. https://doi.org/10.1111/1467-9868.00294	1325 1326 1327 1328
Krishnan V, Eppinger SD, Whitney DE (1997) A model-based framework to overlap product development activities. <i>Management Science</i> 43(4):437–451. https://doi.org/10.1287/mnsc.43.4.437	1329 1330 1331
La Trobe-Bateman J, Wild D (2003) Design for manufacturing: use of a spreadsheet model of manufacturability to optimize product design and development. <i>Research</i>	1332 1333 1334

1335 in Engineering Design 14(2):107–117. <https://doi.org/10.1007/s00163-003-0030-8>
 1336
 1337 Lee JD, See KA (2004) Trust in automation: Designing for appropriate reliance.
 1338 Human Factors: The Journal of the Human Factors and Ergonomics Society
 1339 46(1):50–80. <https://doi.org/10.1518/hfes.46.1.50.30392>
 1340
 1341 Lei J, G’Sell M, Rinaldo A, et al (2018) Distribution-free predictive inference for
 1342 regression. Journal of the American Statistical Association 113(523):1094–1111.
 1343 <https://doi.org/10.1080/01621459.2017.1307116>
 1344
 1345 Leirmo TL (2021) Data on dimensional and geometric accuracy for laser-based powder
 1346 bed fusion of PA12 (PA2200). DataverseNO, version 1.1, <https://doi.org/10.18710/DHACHZ>
 1347
 1348 Lewis KE, Chen W, Schmidt LC (eds) (2006) Decision Making in Engineering Design.
 1349 ASME Press, New York, <https://doi.org/10.1115/1.802469>
 1350
 1351 Madras D, Pitassi T, Zemel R (2018) Predict responsibly: Improving fairness and accu-
 1352 racy by learning to defer. In: Advances in Neural Information Processing Systems,
 1353 vol 31. Curran Associates, Inc., pp 6147–6157, URL [https://proceedings.neurips.](https://proceedings.neurips.cc/paper/2018/hash/09d37c08f7b129e96277388757530c72-Abstract.html)
 1354 [cc/paper/2018/hash/09d37c08f7b129e96277388757530c72-Abstract.html](https://proceedings.neurips.cc/paper/2018/hash/09d37c08f7b129e96277388757530c72-Abstract.html)
 1355
 1356 Malak RJ, Paredis CJJ (2010) Using support vector machines to formalize the
 1357 valid input domain of predictive models in systems design problems. Journal of
 1358 Mechanical Design 132(10):101001. <https://doi.org/10.1115/1.4002151>
 1359
 1360 Montgomery DC (2019) Introduction to Statistical Quality Control, 8th edn. Wiley
 1361
 1362 Oberkampff WL, Roy CJ (2010) Verification and Validation in Scientific
 1363 Computing. Cambridge University Press, Cambridge, [https://doi.org/10.1017/](https://doi.org/10.1017/cbo9780511760396)
 1364 [cbo9780511760396](https://doi.org/10.1017/cbo9780511760396)
 1365
 1366 Pahl G, Beitz W, Feldhusen J, et al (2007) Engineering Design: A Systematic
 1367 Approach, 3rd edn. Springer, London, <https://doi.org/10.1007/978-1-84628-319-2>
 1368
 1369 Rasmussen CE, Williams CKI (2006) Gaussian Processes for Machine Learning. MIT
 1370 Press, Cambridge, MA, <https://doi.org/10.7551/mitpress/3206.001.0001>
 1371
 1372 Regenwetter L, Nobari AH, Ahmed F (2022) Deep generative models in engineering
 1373 design: A review. Journal of Mechanical Design 144(7):071704. [https://doi.org/10.](https://doi.org/10.1115/1.4053859)
 1374 [1115/1.4053859](https://doi.org/10.1115/1.4053859)
 1375
 1376 Roberts DR, Bahn V, Ciuti S, et al (2017) Cross-validation strategies for data with
 1377 temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40(8):913–929.
 1378 <https://doi.org/10.1111/ecog.02881>
 1379
 1380 Romano Y, Patterson E, Candès EJ (2019) Conformalized quantile regression. In:
 Advances in Neural Information Processing Systems, vol 32. Curran Associates,

Inc., pp 3543–3553	1381
	1382
Safdar M, Xie J, Ko H, et al (2024) Transferability analysis of data-driven additive manufacturing knowledge: A case study between powder bed fusion and directed energy deposition. <i>Journal of Computing and Information Science in Engineering</i> 24(5):051010. https://doi.org/10.1115/1.4065090	1383
	1384
	1385
	1386
	1387
Salado A, Kannan H (2018) A mathematical model of verification strategies. <i>Systems Engineering</i> 21(6):593–608. https://doi.org/10.1002/sys.21463	1388
	1389
	1390
Seepersad CC, Pedersen K, Emblemsvåg J, et al (2006) The validation square: How does one verify and validate a design method? In: Lewis KE, Chen W, Schmidt LC (eds) <i>Decision Making in Engineering Design</i> . ASME Press, New York, p 303–313, https://doi.org/10.1115/1.802469.ch25	1391
	1392
	1393
	1394
	1395
Simpson TW, Siddique Z, Jiao JR (eds) (2006) <i>Product Platform and Product Family Design: Methods and Applications</i> . Springer, New York, https://doi.org/10.1007/0-387-29197-0	1396
	1397
	1398
Sobek DK, Ward AC, Liker JK (1999) Toyota’s principles of set-based concurrent engineering. <i>Sloan Management Review</i> 40(2):67–83. URL https://sloanreview.mit.edu/article/toyotas-principles-of-setbased-concurrent-engineering/	1399
	1400
	1401
	1402
Tahera K, Wynn DC, Earl C, et al (2019) Testing in the incremental design and development of complex products. <i>Research in Engineering Design</i> 30(2):291–316. https://doi.org/10.1007/s00163-018-0295-6	1403
	1404
	1405
	1406
Tata M, Thornton AC (1999) Process capability database usage in industry: Myth vs. reality. In: <i>ASME 1999 Design Engineering Technical Conferences, Volume 4: 4th Design for Manufacturing Conference</i> , Las Vegas, Nevada, USA, September 12–16, 1999. American Society of Mechanical Engineers, pp 395–405, https://doi.org/10.1115/detc99/dfm-8968	1407
	1408
	1409
	1410
	1411
	1412
Terwiesch C, Loch CH, De Meyer A (2002) Exchanging preliminary information in concurrent engineering: Alternative coordination strategies. <i>Organization Science</i> 13(4):402–419. https://doi.org/10.1287/orsc.13.4.402.2948	1413
	1414
	1415
	1416
Thomke S, Bell DE (2001) Sequential testing in product development. <i>Management Science</i> 47(2):308–323. https://doi.org/10.1287/mnsc.47.2.308.9838	1417
	1418
	1419
Thomke SH (1998) Simulation, learning and R&D performance: Evidence from automotive development. <i>Research Policy</i> 27(1):55–74. https://doi.org/10.1016/s0048-7333(98)00024-9	1420
	1421
	1422
	1423
Thornton AC (1999) A mathematical framework for the key characteristic process. <i>Research in Engineering Design</i> 11(3):145–157. https://doi.org/10.1007/s001630050011	1424
	1425
	1426

1427 Thornton AC (2004) Variation Risk Management: Focusing Quality Improvements in
 1428 Product Development and Production. Wiley, Hoboken, NJ
 1429
 1430 Thornton AC, Donnelly S, Ertan B (2000) More than just robust design: Why
 1431 product development organizations still contend with variation and its impact on
 1432 quality. *Research in Engineering Design* 12(3):127–143. [https://doi.org/10.1007/
 1433 s001630050028](https://doi.org/10.1007/s001630050028)
 1434
 1435 Tibshirani RJ, Barber RF, Candès EJ, et al (2019) Conformal prediction under covari-
 1436 ate shift. In: *Advances in Neural Information Processing Systems*, vol 32. Curran
 1437 Associates, Inc., pp 2530–2540
 1438
 1439 Wagoner RH, Lim H, Lee MG (2013) Advanced issues in springback. *International*
 1440 *Journal of Plasticity* 45:3–20. <https://doi.org/10.1016/j.ijplas.2012.08.006>
 1441
 1442 Whitney DE (1996) Why mechanical design cannot be like VLSI design. *Research in*
 1443 *Engineering Design* 8(3):125–138. <https://doi.org/10.1007/bf01608348>
 1444
 1445 Williams G, Meisel NA, Simpson TW, et al (2019) Design repository effectiveness for
 1446 3D convolutional neural networks: Application to additive manufacturing. *Journal*
 1447 *of Mechanical Design* 141(11):111701. <https://doi.org/10.1115/1.4044199>
 1448
 1449 Yan X, Wang Z, Zhao C, et al (2026) Machine learning-based cyber manufacturing
 1450 services: A review of manufacturing process selection, process planning, and design
 1451 for manufacturing. *Engineering* 63:47–67. <https://doi.org/10.1016/j.eng.2025.11.034>
 1452
 1453 Yoshida F, Uemori T (2002) A model of large-strain cyclic plasticity describing the
 1454 Bauschinger effect and workhardening stagnation. *International Journal of Plasticity*
 1455 18(5–6):661–686. [https://doi.org/10.1016/S0749-6419\(01\)00050-X](https://doi.org/10.1016/S0749-6419(01)00050-X)
 1456
 1457 Zhang G, Raina A, Cagan J, et al (2021) A cautionary tale about the impact of AI on
 1458 human design teams. *Design Studies* 72:100990. [https://doi.org/10.1016/j.destud.
 1459 2021.100990](https://doi.org/10.1016/j.destud.2021.100990)
 1460
 1461 Zhang Z, Jaiswal P, Rai R (2018) FeatureNet: Machining feature recognition based
 1462 on 3D convolution neural network. *Computer-Aided Design* 101:12–22. [https://doi.
 1463 org/10.1016/j.cad.2018.03.006](https://doi.org/10.1016/j.cad.2018.03.006)
 1464
 1465 Zong Y, Badakhshan S, Zhang J, et al (2026) Trustworthy uncertainty quantifi-
 1466 cation via distributionally-robust dual adaptive conformal prediction. *Journal of*
 1467 *Mechanical Design* 148(2):021703. <https://doi.org/10.1115/1.4069683>
 1468
 1469
 1470
 1471
 1472